Highly Available 3-Tier Web Application on AWS – Implementation Guide

1. Introduction

This project demonstrates how to build a highly available, scalable, and fault-tolerant 3-tier web application on AWS.
The architecture ensures high availability, auto scaling, secure HTTPS communication, and durable storage.

2. Objectives

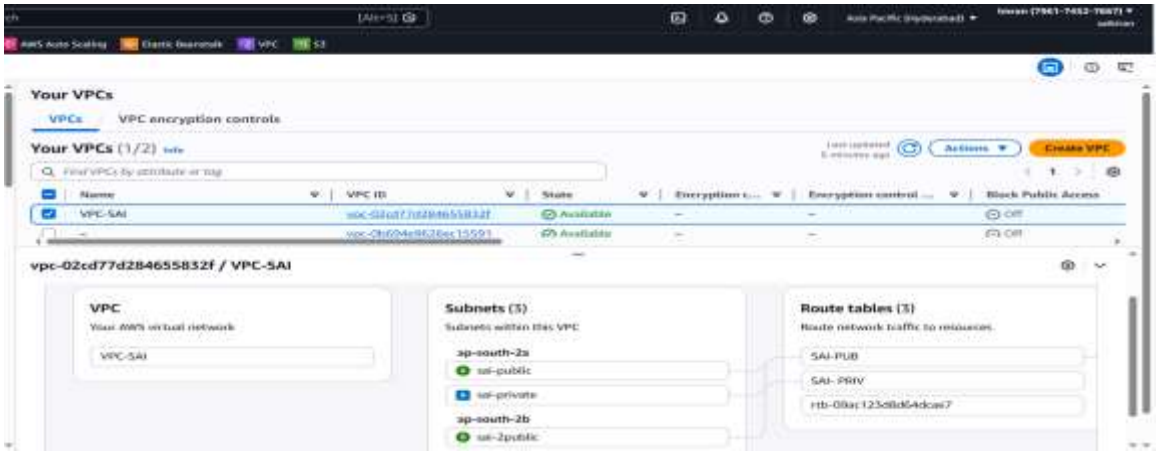
- Deploy 3-tier architecture
- Configure Auto Scaling
- Implement Load Balancer
- Secure using HTTPS
- Setup storage and backups

3. Architecture Overview

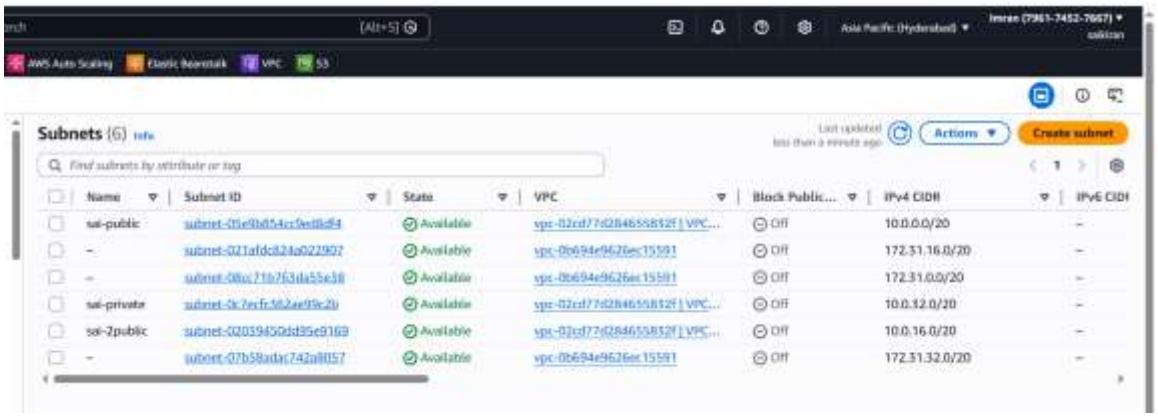
Internet → ALB (HTTPS) → EC2 (ASG) → EBS → S3

4. VPC & Networking Setup

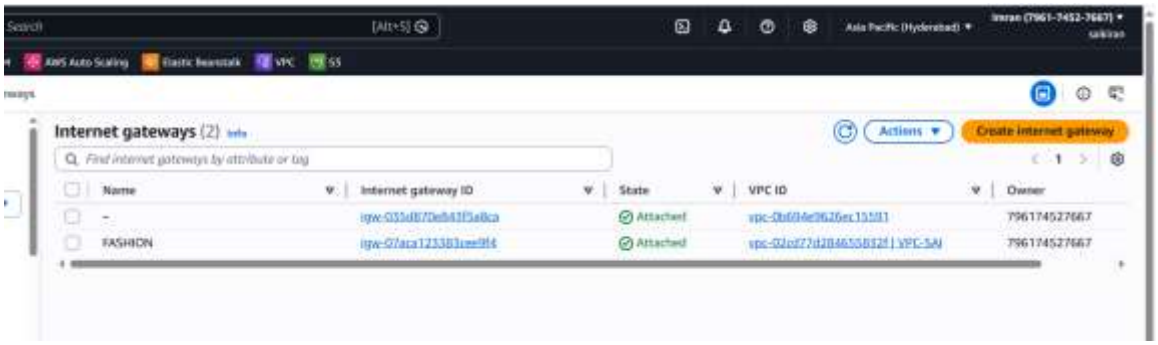
Create VPC (10.0.0.0/16)



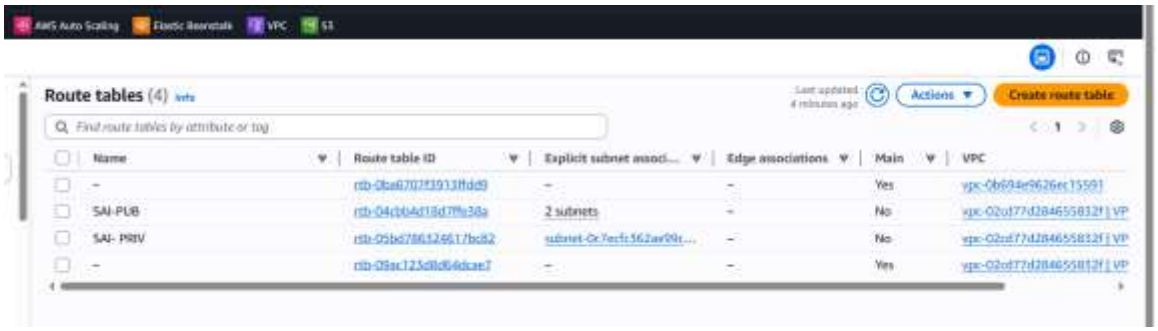
Create Public & Private Subnets



Attach Internet Gateway



Configure Route Tables



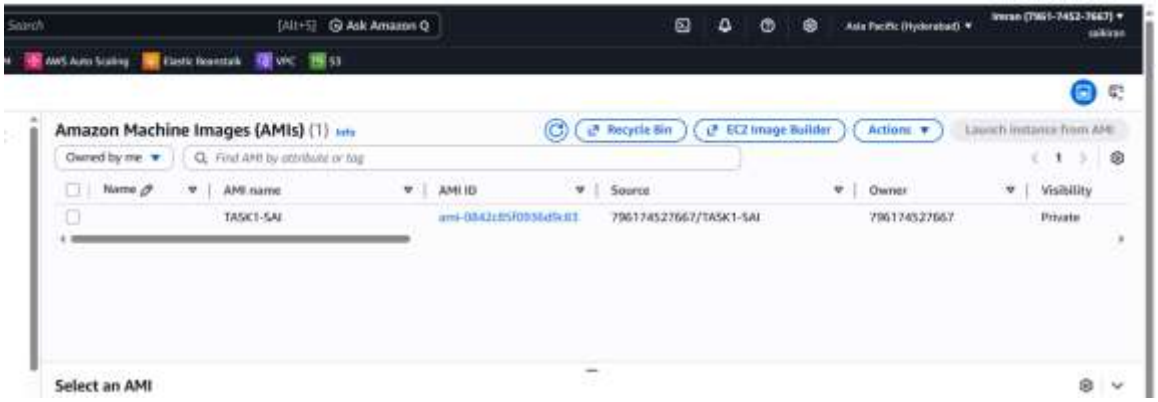
5. Security Configuration

Setup Security Groups:

- ALB: Allow 80, 443
- EC2: Allow ALB traffic
- Bastion: Allow SSH

6. EC2 & AMI Setup

Launch EC2 → Install Nginx/Apache → Create Golden AMI

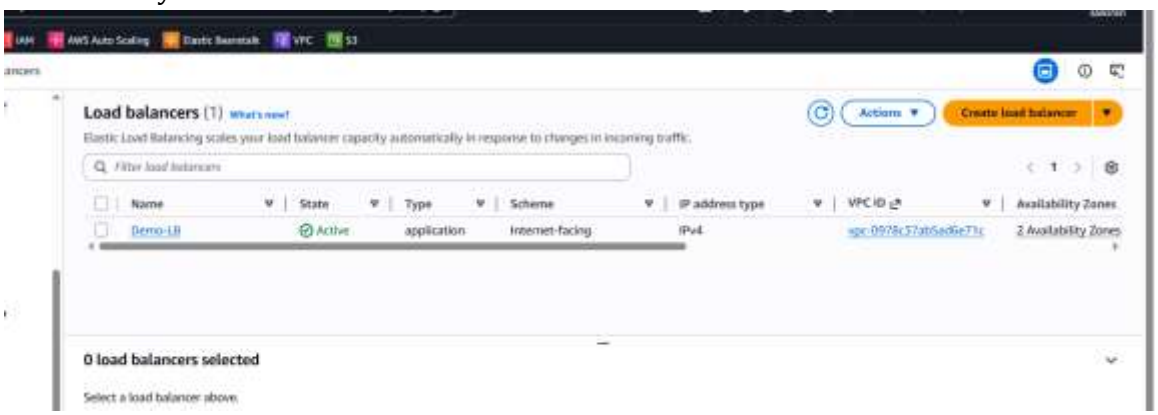


Configure User data in EC2 while launching

7. Load Balancer Setup

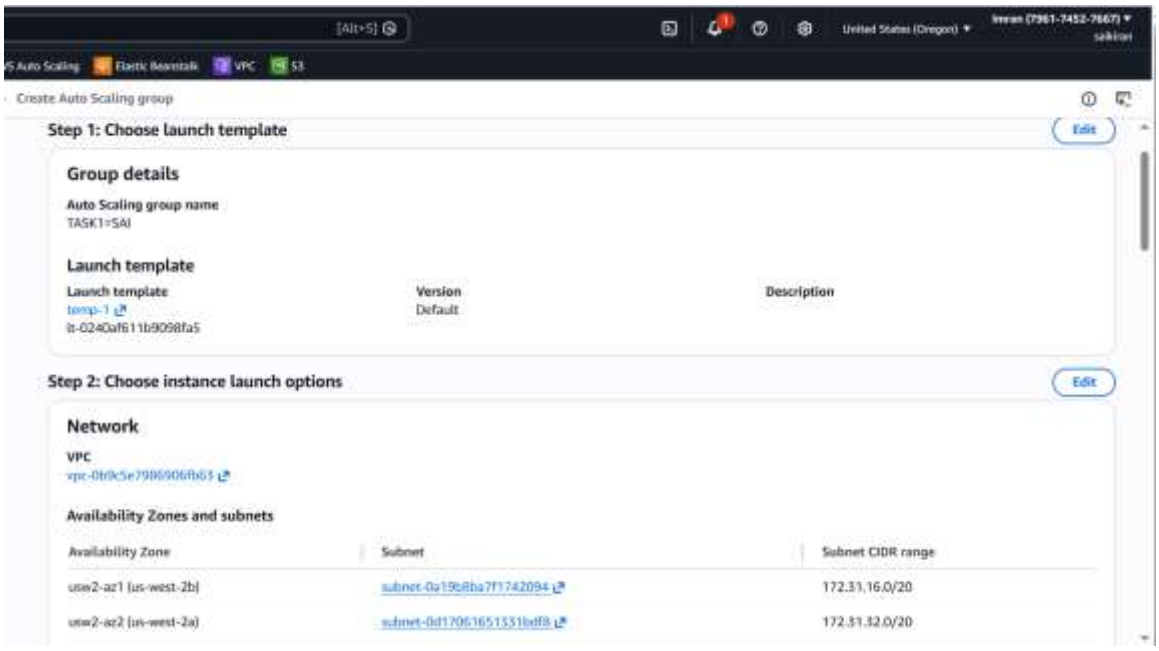
Create ALB
Configure HTTP→HTTPS redirect
Enable Cross-zone LB

Enable Sticky Sessions



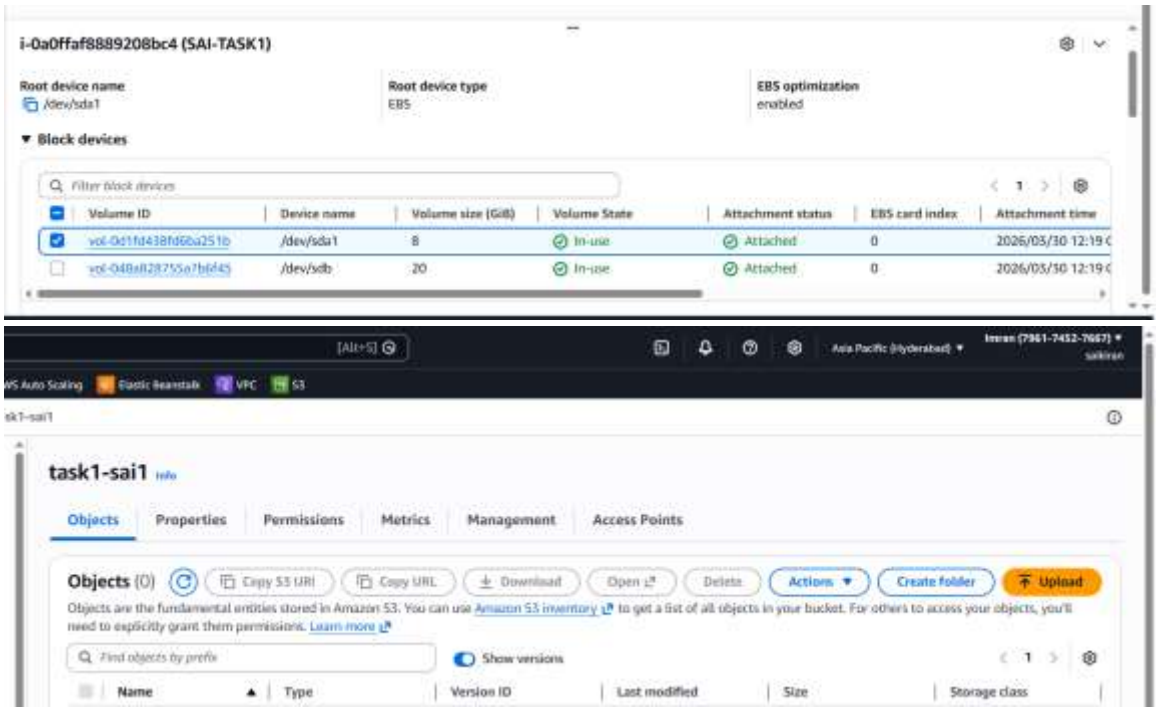
8. Auto Scaling Group

Launch Template using AMI
Min=2, Max=6
Scaling: CPU >60% scale out, <30% scale in
Cooldown: 300s



9. Storage Configuration

EBS: 20GB gp3 encrypted
S3: Versioning enabled, SSE-S3
AWS Backup: Daily snapshots (7 days retention)



10. Validation

Check ALB health
Perform stress test
Verify HTTPS redirect
Check S3 asset loading
Confirm backup snapshots

11. Conclusion

This project demonstrates a production-ready scalable AWS architecture.